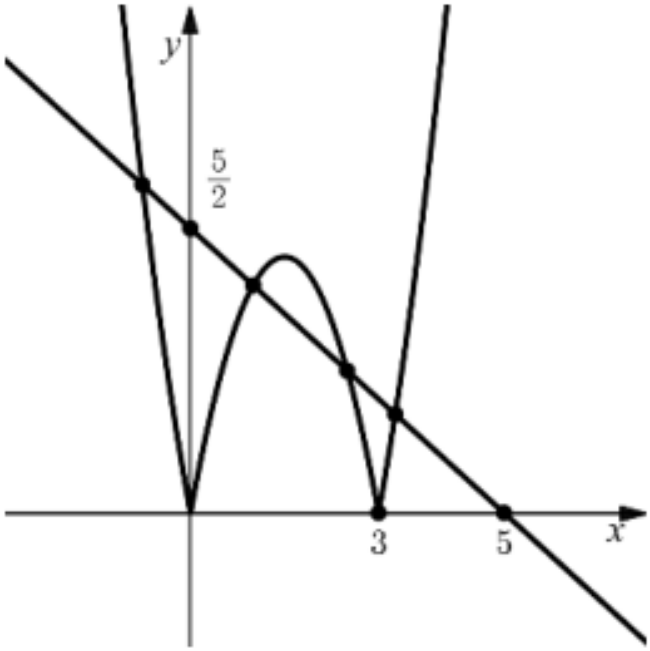
2.57 Modulus of Functions: Inequalities

Worked Solutions

1 a)

Sketch the curves $y = |x^2 - 3x|$ and $y = \frac{5}{2} - \frac{1}{2}x$ on the same set of axes.



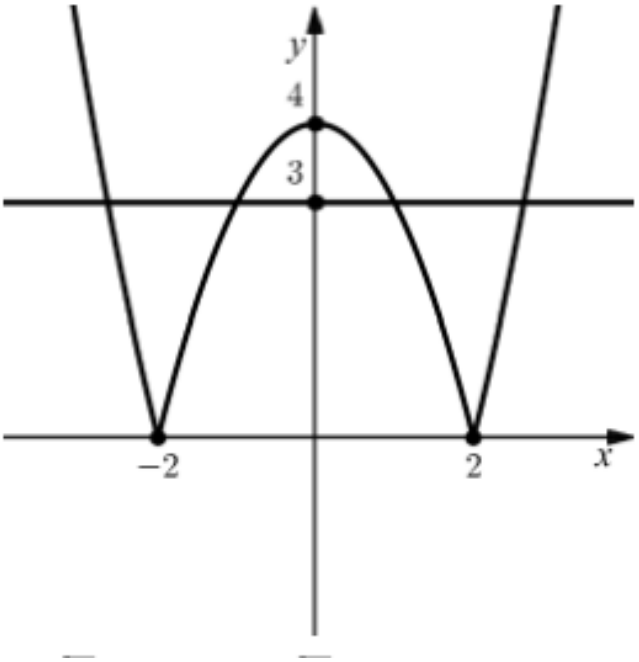
1 b)

$$x^2 - 3x = \frac{5}{2} - \frac{1}{2}x \Rightarrow 2x^2 - 5x - 5 = 0 \Rightarrow x = \frac{5}{4} \pm \frac{\sqrt{65}}{4}$$

$$x^2 - 3x = -\frac{5}{2} + \frac{1}{2}x \Rightarrow 2x^2 - 7x + 5 = 0 \Rightarrow x = 1 \text{ or } x = \frac{5}{2}$$

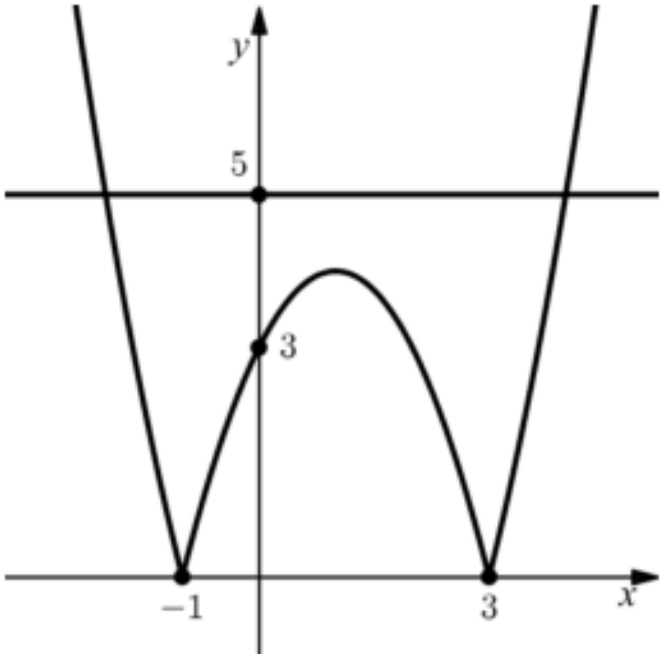
$$\therefore x \in \left(-\infty, \frac{5}{4} - \frac{\sqrt{65}}{4}\right) \cup \left(1, \frac{5}{2}\right) \cup \left(\frac{5}{4} + \frac{\sqrt{65}}{4}, \infty\right).$$

2 a)



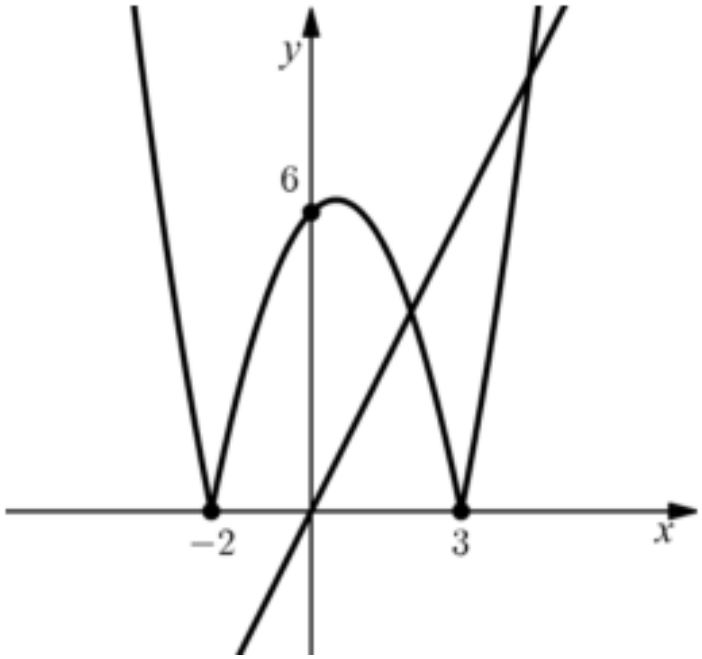
$$x^2 - 4 = 3 \Rightarrow x^2 = 7 \Rightarrow x = -\sqrt{7} \text{ or } x = \sqrt{7}$$
$$x^2 - 4 = -3 \Rightarrow x^2 = 1 \Rightarrow x = -1 \text{ or } x = 1$$
$$\therefore \{x : x \in \mathbb{R},\ x < -\sqrt{7}\} \cup \{x : x \in \mathbb{R},\ -1 < x < 1\} \cup \{x : x \in \mathbb{R},\ x > \sqrt{7}\}.$$

2 b)



$$x^2 - 2x - 3 = 5 \Rightarrow x^2 - 2x - 8 = 0 \Rightarrow x = -2 \text{ or } x = 4.$$
$$\therefore \{x : x \in \mathbb{R},\ -2 \leq x \leq 4\}.$$

2 c)



$$-(x^2 - x - 6) = 2x \Rightarrow x^2 + x - 6 = 0 \Rightarrow x = -3 \text{ or } x = 2$$

$$x^2 - x - 6 = 2x \Rightarrow x^2 - 3x - 6 = 0 \Rightarrow x = \frac{3}{2} \pm \frac{\sqrt{33}}{2}$$

$$\therefore \{x : x \in \mathbb{R},\ x < 2\} \cup \left\{x : x \in \mathbb{R},\ x > \frac{3}{2} + \frac{\sqrt{33}}{2}\right\}.$$

3 a)

$$k(x+3)(x-2), \quad k = -2.$$

$$\pm 2(x+3)(x-2) = \pm(2x^2 + 2x - 12).$$

$$\therefore a = 2,\ b = 2,\ c = -12 \quad \text{or} \quad a = -2,\ b = -2,\ c = 12.$$

3 b)

$$2x^2 + 2x - 12 = x \Rightarrow 2x^2 + x - 12 = 0 \Rightarrow x = -\frac{1}{4} \pm \frac{\sqrt{97}}{4}$$

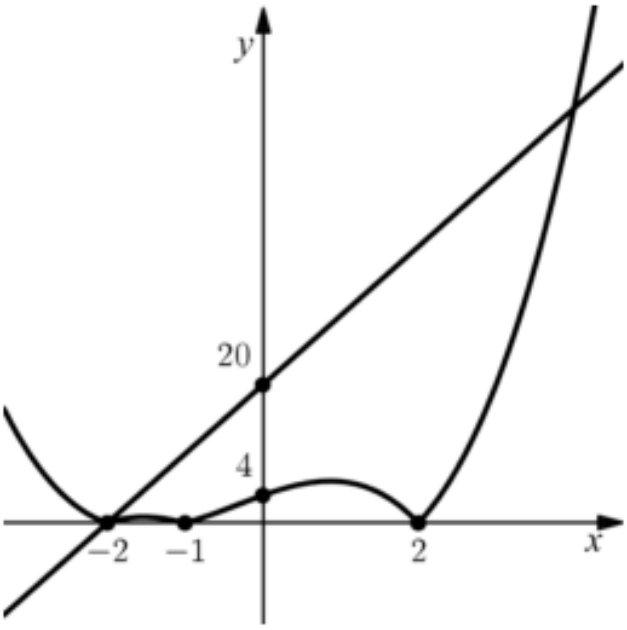
$$2x^2 + 2x - 12 = -x \Rightarrow 2x^2 + 3x - 12 = 0 \Rightarrow x = -\frac{3}{4} \pm \frac{\sqrt{105}}{4}$$

$$\therefore x = -\frac{1}{4} + \frac{\sqrt{97}}{4} \quad \text{and} \quad x = -\frac{3}{4} + \frac{\sqrt{105}}{4}$$

$$\left\{x : x \in \mathbb{R},\ -\frac{3}{4} + \frac{\sqrt{105}}{4} < x < -\frac{1}{4} + \frac{\sqrt{97}}{4}\right\}.$$

4 a)

Sketch the curves $y = |x^3 + x^2 - 4x - 4|$ and $y = 10x + 20$ on the same set of axes.



4 b)

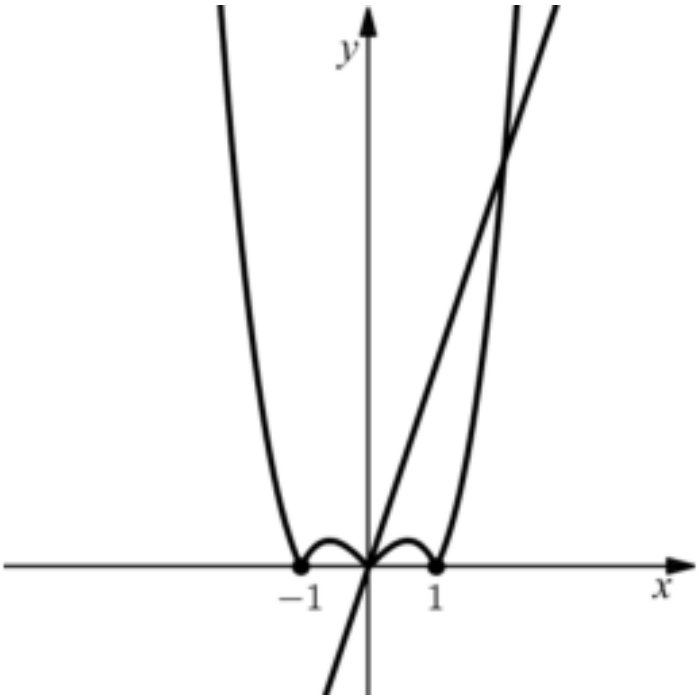
$$x^3 + x^2 - 4x - 4 = 10x + 20 \Rightarrow x^3 + x^2 - 14x - 24 = 0$$
$$\Rightarrow (x - 4)(x + 2)(x + 3) = 0$$
$$x = -3 \text{ or } x = -2 \text{ or } x = 4$$

So $x = -2$ or $x = 4$.

$$\therefore x \in (-\infty, -2) \cup (4, \infty).$$

5 a)

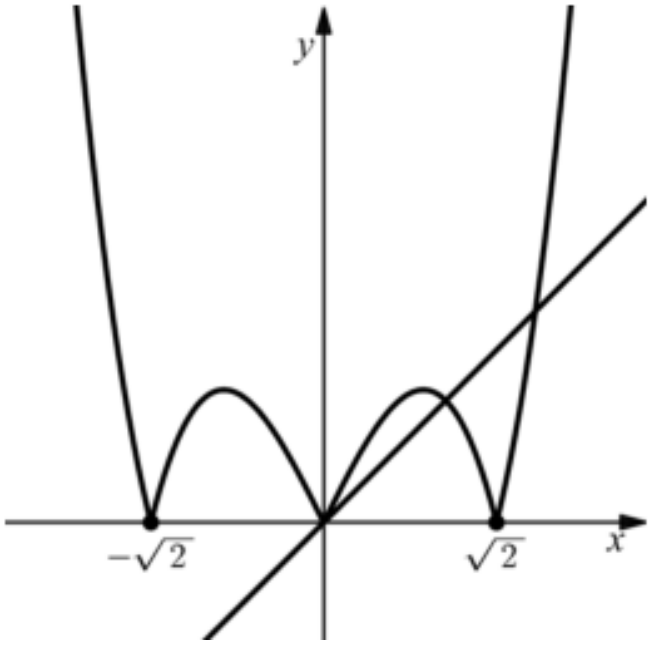
$$x^3 - x = 3x \Rightarrow x^3 - 4x = 0 \Rightarrow x(x - 2)(x + 2) = 0 \Rightarrow x = -2,\ x = 0 \text{ or } x = 2$$
$$x^3 - x = -3x \Rightarrow x^3 + 2x = 1 \Rightarrow x(x^2 + 2) = 0 \Rightarrow x = 0$$



$\therefore \{x : x \in \mathbb{R},\ x < 0\} \cup \{x : x \in \mathbb{R},\ x > 2\}.$

5 b)

$$x^3 - 2x = x \Rightarrow x^3 - 3x = 0 \Rightarrow x(x^2 - 3) = 0 \Rightarrow x = 0 \text{ or } x = \pm\sqrt{3}$$
$$x^3 - 2x = -x \Rightarrow x^3 - x = 0 \Rightarrow x(x^2 - 1) = 0 \Rightarrow x = 0 \text{ or } x = \pm 1$$



$\therefore \{x : x \in \mathbb{R},\ x < 0\} \cup \{x : x \in \mathbb{R},\ 0 < x < 1\} \cup \{x : x \in \mathbb{R},\ x > \sqrt{3}\}.$

5 c)

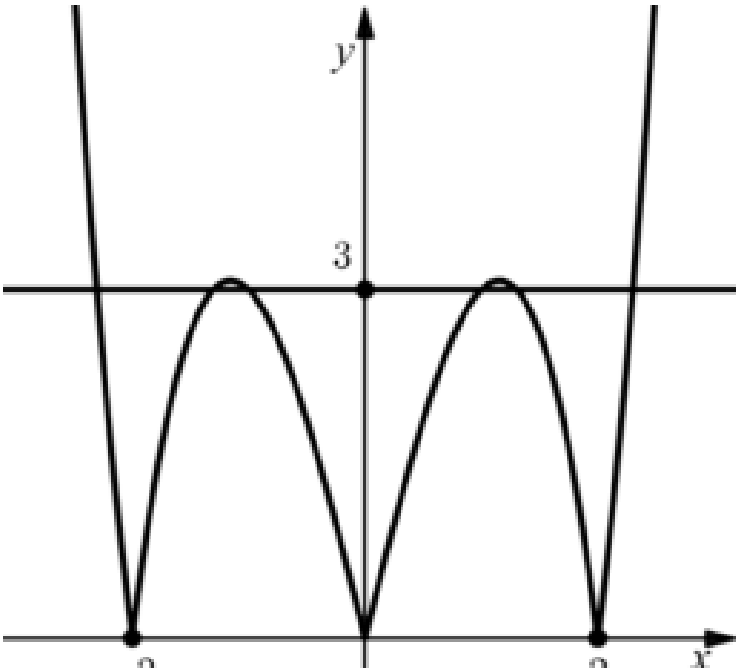
$$x^3 - 4x = x(x^2 - 4) = x(x-2)(x+2)$$

$$x^3 - 4x = 3 \Rightarrow x^3 - 4x - 3 = 0 \Rightarrow (x+1)(x^2 - x - 3) = 0$$

$$x = -1 \text{ or } x = \frac{1}{2} \pm \frac{\sqrt{13}}{2}$$

$$x^3 - 4x = -3 \Rightarrow x^3 - 4x + 3 = 0 \Rightarrow (x-1)(x^2 + x - 3) = 0$$

$$x = 1 \text{ or } x = -\frac{1}{2} \pm \frac{\sqrt{13}}{2}$$



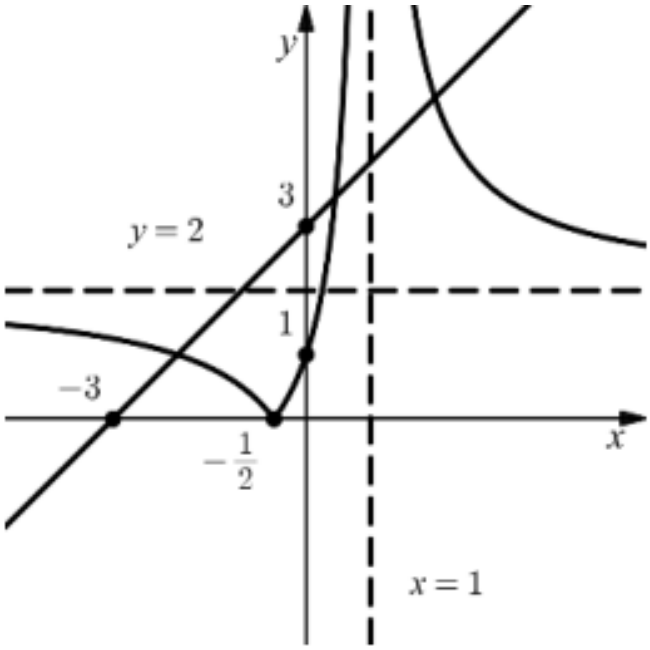
$$\therefore \left\{ x : x \in \mathbb{R},\ -\frac{1}{2} - \frac{\sqrt{13}}{2} \leq x \leq \frac{1}{2} - \frac{\sqrt{13}}{2} \right\}$$

$$\cup \{ x : x \in \mathbb{R},\ -1 \leq x \leq 1 \}$$

$$\cup \left\{ x : x \in \mathbb{R},\ -\frac{1}{2} + \frac{\sqrt{13}}{2} \leq x \leq \frac{1}{2} + \frac{\sqrt{13}}{2} \right\}.$$

6 a)

Consider $y = \dfrac{2x+1}{x-1}$. When $x = 0$, $y = -1$; when $y = 0$, $x = -\frac{1}{2}$. Asymptotes: $x = 1$, $y = 2$. Sketch $y = \left| \dfrac{2x+1}{x-1} \right|$ and $y = x + 3$.



6 b)

$$\frac{2x+1}{x-1} = x+3 \Rightarrow 2x+1 = (x+3)(x-1) \Rightarrow x^2-4=0 \Rightarrow x=-2 \text{ or } x=2$$

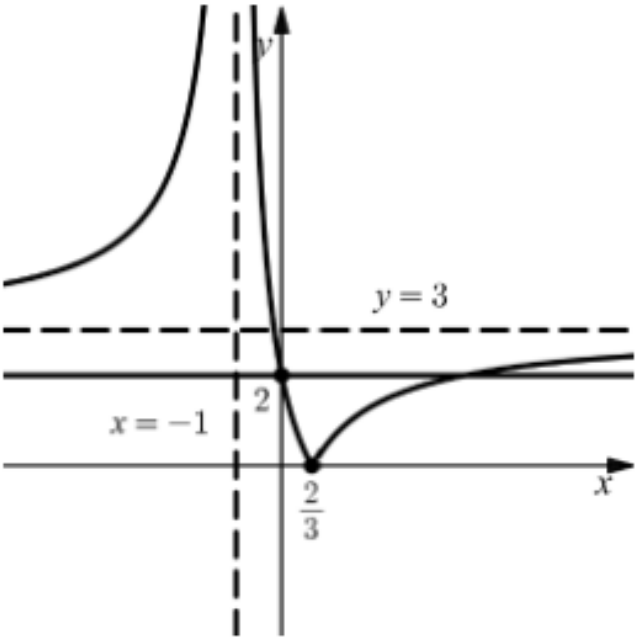
$$\frac{2x+1}{x-1} = -(x+3) \Rightarrow 2x+1 = -(x+3)(x-1) \Rightarrow x^2+4x-2=0 \Rightarrow x=-2\pm\sqrt{6}$$

So $x=-2+\sqrt{6}$.

$$\therefore \{x : x\in\mathbb{R},\ x<-2\} \cup \{x : x\in\mathbb{R},\ -2+\sqrt{6}<x<1\} \cup \{x : x\in\mathbb{R},\ 1<x<2\}.$$

7 a)

Consider $y=\dfrac{3x-2}{x+1}$. When $x=0$, $y=-2$; when $y=0$, $x=\frac{2}{3}$. Asymptotes: $x=-1$, $y=3$. Sketch $y=\left|\dfrac{3x-2}{x+1}\right|$ and $y=2$.



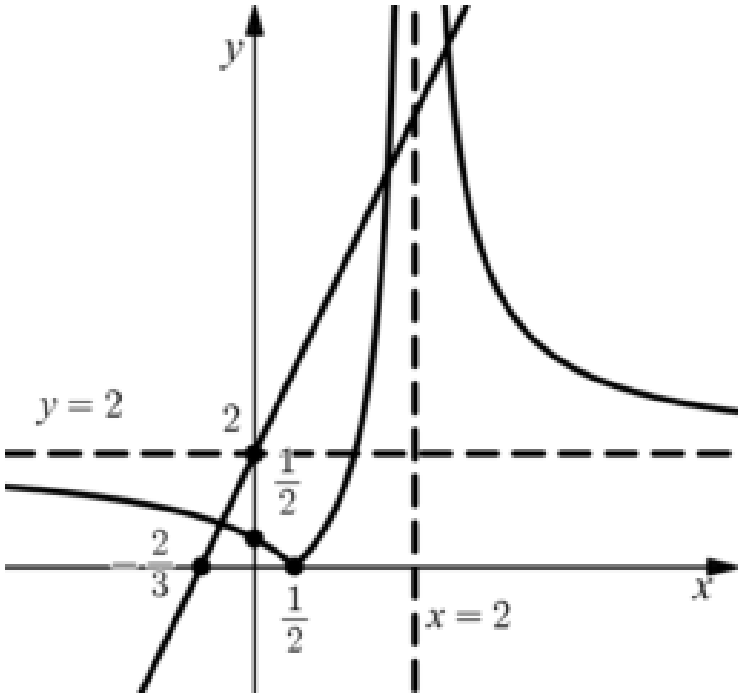
1) → x ≠ 4

$$\frac{3x-2}{x+1} = 2 \Rightarrow 3x-2 = 2(x+1) \Rightarrow x = 4$$

$$\therefore \{x : x \in \mathbb{R},\ x < -1\} \cup \{x : x \in \mathbb{R},\ -1 < x < 0\} \cup \{x : x \in \mathbb{R},\ x > 4\}.$$

7 b)

Consider $y = \dfrac{2x-1}{x-2}$. When $x = 0$, $y = \frac{1}{2}$; when $y = 0$, $x = \frac{1}{2}$. Asymptotes: $x = 2$, $y = 2$. Sketch $y = \left|\dfrac{2x-1}{x-2}\right|$ and $y = 3x + 2$.



$$\frac{2x-1}{x-2} = 3x+2 \Rightarrow 2x-1 = (3x+2)(x-2) \Rightarrow 3x^2 - 6x - 3 = 0$$

$$\Rightarrow x = 1 + \sqrt{2} \text{ or } x = 1 - \sqrt{2}$$

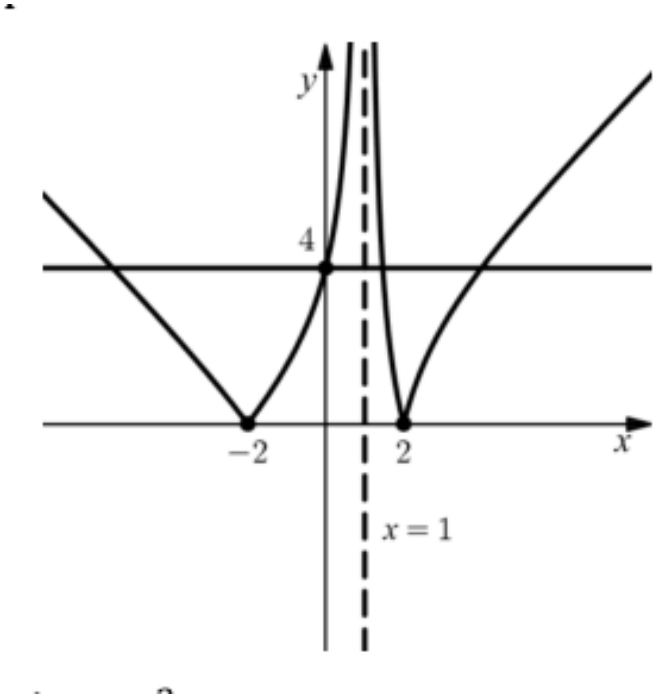
$$\frac{2x-1}{x-2} = -(3x+2) \Rightarrow 2x - 1 = -(3x+2)(x-2) \Rightarrow 3x^2 - 2x - 5 = 0$$

$$\Rightarrow x = -1 \text{ or } x = \frac{5}{3}, \quad \text{so } x = \frac{5}{3}$$

$$\therefore \{x : x \in \mathbb{R},\ 1 - \sqrt{2} < x < \frac{5}{3}\} \cup \{x : x \in \mathbb{R},\ x > 1 + \sqrt{2}\}.$$

7 c)

Consider $y = \dfrac{x^2-4}{x-1}$. When $x = 0$, $y = 4$; when $y = 0$, $x = \pm 2$. Asymptote: $x = 1$. Sketch $y = \left|\dfrac{x^2-4}{x-1}\right|$ and $y = 4$.

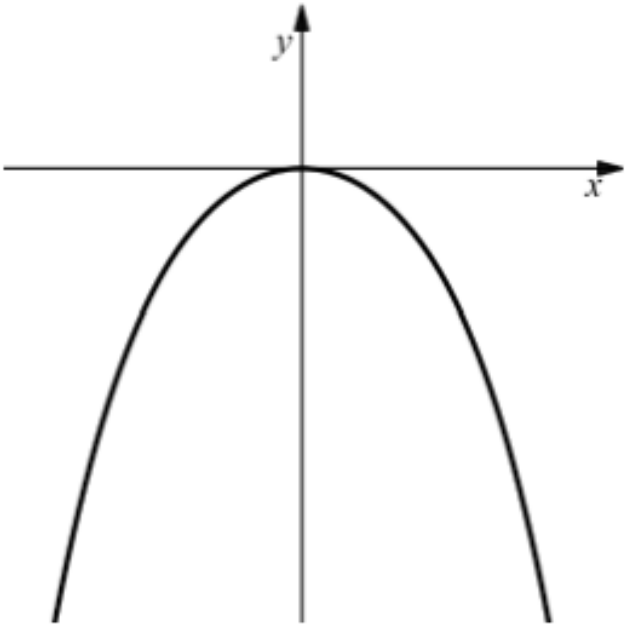


$$\frac{x^2-4}{x-1} = 4 \Rightarrow x^2 - 4 = 4(x-1) \Rightarrow x^2 - 4x = 0 \Rightarrow x = 0 \text{ or } x = 4$$

$$\frac{x^2-4}{x-1} = -4 \Rightarrow x^2 - 4 = -4(x-1) \Rightarrow x^2 + 4x - 8 = 0 \Rightarrow x = -2 \pm 2\sqrt{3}$$

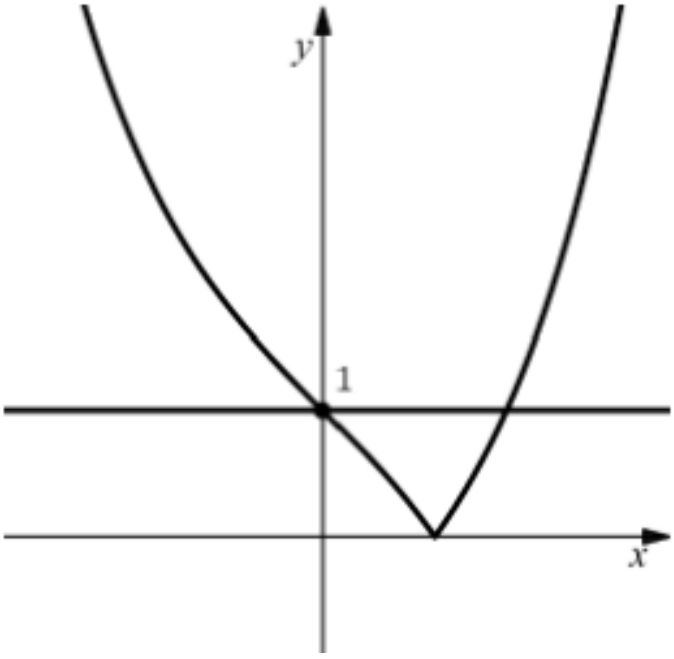
$$\therefore \{x : x \in \mathbb{R},\ -2 - 2\sqrt{3} \leq x \leq 0\} \cup \{x : x \in \mathbb{R},\ -2 + 2\sqrt{3} \leq x \leq 4\}.$$

8 a)



$$x \in (-\infty, 0) \cup (0, \infty).$$

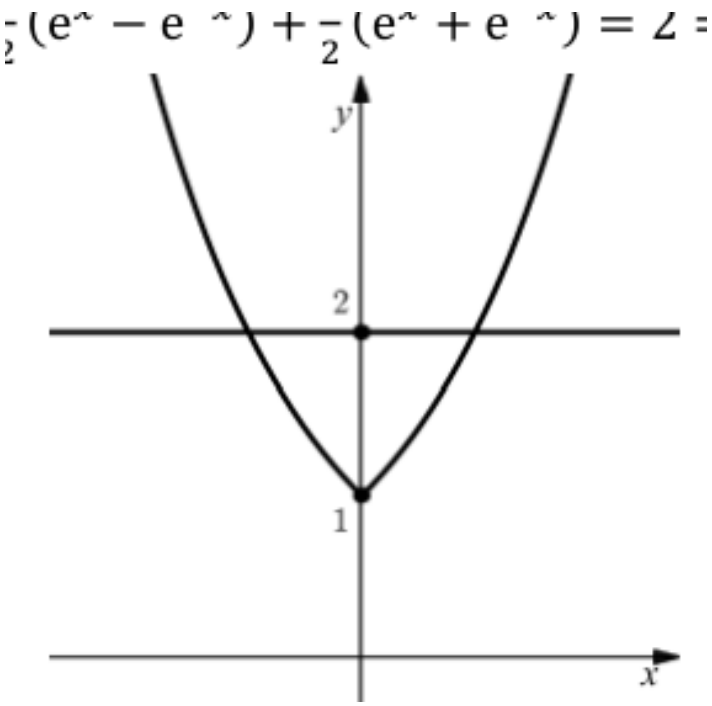
8 b)



$$1 - \sinh x = 1 \Rightarrow \sinh x = 0 \Rightarrow x = \ln(0 + \sqrt{0^2 + 1}) = 0$$
$$1 - \sinh x = -1 \Rightarrow \sinh x = 2 \Rightarrow x = \ln(2 + \sqrt{2^2 + 1}) = \ln(2 + \sqrt{5})$$

$$x \in [-1, \ln(2 + \sqrt{5})].$$

8 c)



$$\sinh x + \cosh x = 2 \Rightarrow \frac{1}{2}(e^x - e^{-x}) + \frac{1}{2}(e^x + e^{-x}) = 2 \Rightarrow e^x = 2 \Rightarrow x = \ln 2$$

$$-\sinh x + \cosh x = 2 \Rightarrow -\frac{1}{2}(e^x - e^{-x}) + \frac{1}{2}(e^x + e^{-x}) = 2 \Rightarrow e^{-x} = 2$$

Then by symmetry, $x = -\ln 2$.

$$\therefore x \in (-\infty, -\ln 2] \cup [\ln 2, \infty).$$

9)

$$|-1 + \ln x| \leq 2.$$

$$-1 + \ln x = 2 \Rightarrow \ln x = 3 \Rightarrow x = e^3$$
$$-1 + \ln x = -2 \Rightarrow \ln x = -1 \Rightarrow x = e^{-1}$$

$$\therefore \{x : x \in \mathbb{R},\ e^{-1} \leq x \leq e^3\}.$$

10)

$$\left|\frac{1}{x}\right| \geq 1.$$

$$\frac{1}{x} = 1 \Rightarrow x = 1$$
$$\frac{1}{x} = -1 \Rightarrow x = -1$$

$$\therefore \{x : x \in \mathbb{R},\ -1 \leq x < 0\} \cup \{x : x \in \mathbb{R},\ 0 < x \leq 1\}.$$

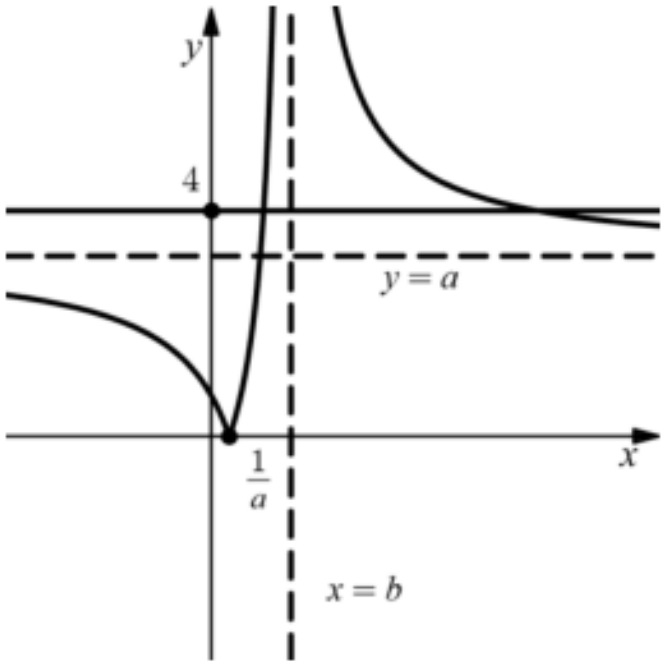
11)

$$\left|\cos x - \frac{1}{2}\right| \geq \frac{1}{2}.$$

$$\cos x - \frac{1}{2} = \frac{1}{2} \Rightarrow \cos x = 1 \Rightarrow x = 0$$
$$\cos x - \frac{1}{2} = -\frac{1}{2} \Rightarrow \cos x = 0 \Rightarrow x = -\frac{\pi}{2} \text{ or } x = \frac{\pi}{2}$$
$$x \in \left(-\pi, -\frac{\pi}{2}\right] \cup [0] \cup \left[\frac{\pi}{2}, \pi\right].$$

12)

Consider $y = \dfrac{ax-1}{x-b}$. When $x = 0$, $y = \frac{1}{b}$; when $y = 0$, $x = \frac{1}{a}$. Asymptotes: $x = b$, $y = a$. Sketch $y = \left|\dfrac{ax-1}{x-b}\right|$ and $y = 4$.



$$\frac{ax-1}{x-b} = 4 \Rightarrow ax - 1 = 4(x-b) \Rightarrow ax - 1 = 4x - 4b \Rightarrow ax - 4x = 1 - 4b$$
$$\Rightarrow x = \frac{1-4b}{a-4}$$
$$\frac{ax-1}{x-b} = -4 \Rightarrow ax - 1 = -4(x-b) \Rightarrow ax - 1 = 4b - 4x \Rightarrow ax + 4x = 1 + 4b$$
$$\Rightarrow x = \frac{1+4b}{a+4}$$
$$\therefore \left\{x : x \in \mathbb{R},\ \frac{1+4b}{a+4} \leq x < b\right\} \cup \left\{x : x \in \mathbb{R},\ b < x \leq \frac{1-4b}{a-4}\right\}.$$